

CO 5	✓	✓		✓
CO 6	✓			

Programme	BSc Statistics				
Course Code	STA3MN211 (P)				
Course Title	Probability theory and statistical distributions				
Type of Course	Minor				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Familiarity with basic calculus such as differentiation and integration, basic knowledge of set theory. Experience with basic data visualization techniques.				
Course Summary	Provide students with a solid foundation in probability theory, including classical and axiomatic approaches, conditional probability, random variables, probability distributions and their applications.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Calculate probabilities of events using classical probability rules and understand their limitations.	U	C	Instructor-created exams / Quiz
CO2	Determine marginal probabilities and identify their role in joint probability distributions.	R	C	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Define random variables and distinguish between discrete and continuous random variables and analyze data to help entrepreneurial decisions using critical thinking skills.	R	F	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Understand the significance of probability distributions in statistical analysis and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	Instructor-created exams / Home Assignments
CO5	Apply discrete and continuous probability distributions and understand their properties and applications.	Ap	F	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Perform computations involving probabilities and using R programming language.	Ap	P	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Probability		9	15
	1	Basic concepts of Probability ,Classical definition of Probability ,Axiomatic approach to Probability	2	
	2	Addition Theorem, Multiplication Theorem	3	
	3	Conditional Probability	2	
	4	Independence of events	2	
	Sections from References: Unit 1: 3.3-3.8 [Ref 1] Unit 2: 3.9,3.11 [Ref 1] Unit 3: 3.10[Ref 1] Unit 4: 3.11-3.15 [Ref 1]			
II	Random Variables		9	15
	5	Random Variables, Discrete and continuous random variables	2	
	6	Probability distribution , Distribution function (Applications in discrete case)	3	
	7	Mathematical expectation (Applications in discrete case)	2	
	8	Variance (Applications in discrete case)	2	
	Sections from References: 4. Unit 5&6: 5.1-5.4.2 [Ref 1] 5. Unit 7: 6.1-6.4 [Ref 1] 6. Unit 8: 6.6 [Ref 1]			
III	Discrete and Continuous distributions		19	25
	9	Binomial distribution (Definition and problems)	2	
	10	Poisson distribution (Definition and problems)	2	
	11	Normal distribution (Definition and problems)	1	
	12	Properties of Normal distribution	3	
	13	Uniform distribution (Definition and properties)	3	
	14	Exponential distribution (Definition and properties)	3	
	15	Gamma distribution (Definition and properties)	1	
	16	Beta distribution (Definition and properties)	1	
	17	Cauchy, Pareto distribution (Definition only)	3	
	Sections from References: Unit 9:14.2 [Ref 2] Unit 10:14.3[Ref 2] Unit 11:14.4 [Ref 2] Unit 12:14.5 [Ref 2] Unit 13:14.6[Ref 2] Unit 14:14.7 [Ref 2] Unit 15:14.8 [Ref 2] Unit 16&17:14.9[Ref 2]			
IV	R programming		8	15
	18	R as a set of statistical tables	2	

	19	cumulative distribution	2	
	20	probability density function	2	
	21	plotting probability curves for standard discrete distributions.	1	
	22	plotting probability curves for standard continuous distributions	1	
	Sections from References: Unit 18: 3.3 [Ref 3] Unit 19: 3.3 [Ref 3] Unit 20: 3.3 [Ref 3] Unit 21:3.4&3.5 [Ref 3] Unit 22: 3.4 [Ref 3]			
V	PRACTICUM		30	
		Do practice problems in R software from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course. 1. Graph of Binomial distribution 2. Graph of Poisson distribution 3. Graph of Normal distribution 4. Graph of Uniform distribution 5. Graph of Exponential distribution 6. Graph of Gamma distribution 7. Graph of Beta distribution 8. Graph of Cauchy distribution		
	Sections from References: Unit 1,2: 3.3 [Ref 3] Unit 3,4,5: 3.4 [Ref 3] Unit 6,7,8: 3.5 [Ref 3]			
Books and References: 1. Gupta, S. C. and Kapoor, V. K. (2020). Fundamentals of Mathematical Statistics, 12 th edition, Sulthan Chand, New Delhi 2. Gupta, S. C.. (2015). Fundamentals of Statistics, Himalaya Publishing House 3. Sudha G Purohith, Sharad D Core, Shailaja R Deshmukh ,Statistics Using R(2015)				

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	3	2	-	-	-	-	1	-	-	-	-	-
CO 2	-	-	-	-	1	-	-	1	-	1	-	-
CO 3	-	-	-	2	-	-	-	-	-	-	3	
CO 4	-	-	-	-	-	1	-	-	-	-	-	2
CO 5	-	3	-	1	-	-	-	-	2	-	-	-
CO 6	-	-	1	-	2	3	-	-	-	-	3	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignm ent	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5	✓	✓		✓

CO 6	✓			
------	---	--	--	--

MINOR COURSES IN ACTUARIAL SCIENCE **SYLLABUS**